

Full Training and Runtime Pipeline

Direct temporal token-to-BEV decoding with frozen live VGGT- Ω features, independent observed and complete-evidential models, learned non-metric VGGT-normalized support, and separate single-frame and merged outputs.

Central routing rule

Merged BEV is decoded directly from the complete RGB/VGGT temporal feature history.

Predicted single-frame BEVs are never used as inputs to the merged decoder.

RGB-only runtime

Frozen live VGGT- Ω

Direct Method I head

Non-metric scale

512²

single-frame BEV

800²

merged temporal BEV

1×

VGGT forward per RGB batch

Project:

/home/wolfie/Project/VGGTBEV_Method1

Reference: bev_head_methods.pdf — Method I

Primary production mode: Model B only

Checkpoint contract: format version 10

Document date: 28 July 2026

Coordinate system: VGGT scene-radius normalized

1. Contract at a Glance

Runtime input
images: float tensor [B, N, 3, 384, 512]

- Contiguous RGB history ending at the current frame
- No camera height
- No GT depth, intrinsics, extrinsics, trajectory, or BEV

Runtime outputs

- Single BEV: **512 × 512**
- Merged BEV: **800 × 800**
- Learned normalized extent and cell size per branch/sample
- Optional live VGGT diagnostic geometry

What this implementation is

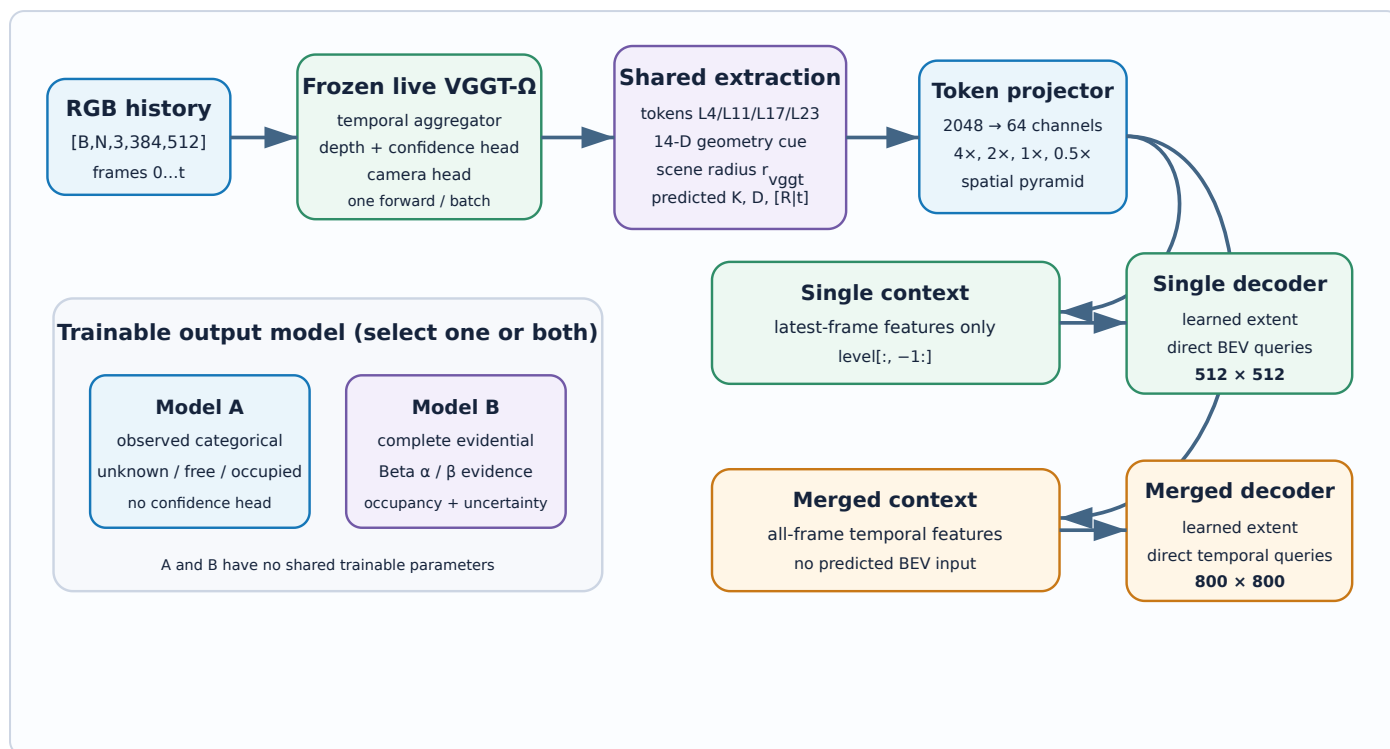
Property	Current implementation	Consequence
Method family	PDF Method I: direct token-to-BEV	No lift, splat, or geometric rasterization
Backbone	Live frozen VGGT-Ω, evaluated during every forward	Training and runtime see the same predicted geometry
Temporal merged path	All RGB frames → one VGGT temporal extraction → merged decoder	No cascade through predicted single BEVs
Single path	Latest-frame projected VGGT features only	Current-view local output
Scale	Learned support in scene-radius-normalized VGGT units	Not metres; no fixed 6.5/10 output range
Model isolation	Model A and B share no trainable parameters	Losses cannot damage the other model

Source labels versus model coordinates. Training reads the available simulator targets covering 6.5×6.5 m (single) and 10×10 m (merged). Those values describe source supervision and metric evaluation. They are not passed into `model.forward` and do not fix the output range.

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2. End-to-End Architecture



■ model/data flow ■ frozen extraction ■ merged temporal path ■ evidential path

This remains Method I because image tokens are decoded directly by learned BEV queries. VGGT depth and pose contribute a normalized geometry summary, but no reconstructed image feature is assigned or splatted into a BEV cell.

3. Training Data and Preprocessing

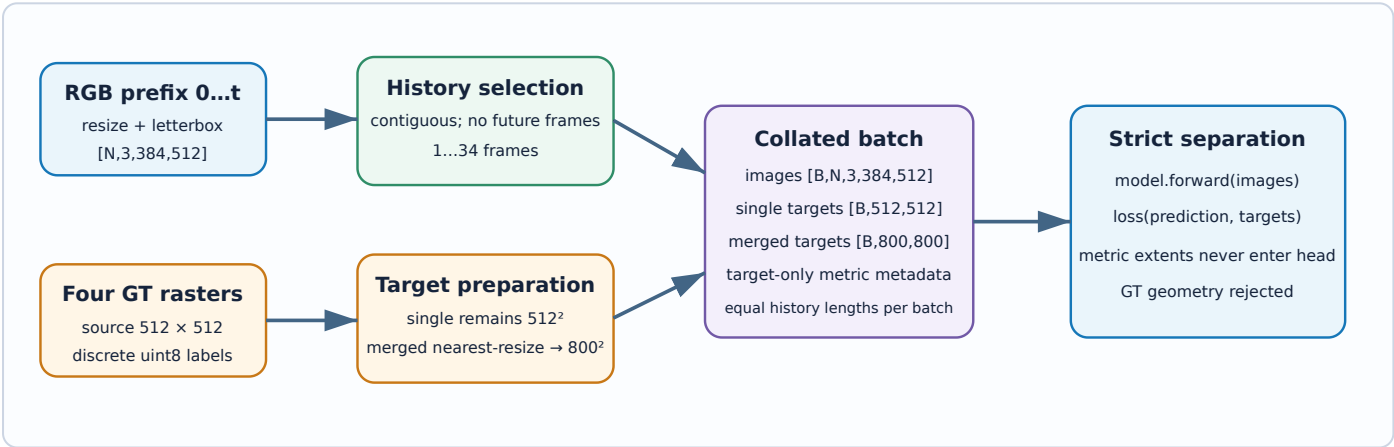
Per-session source files

Source	Use
camera/frame_*.png	Only model input
bev_6p5m/masked	Single observed target
bev_6p5m/complete	Single complete target
merged_masked_10m	Merged observed target
merged_complete_10m	Merged complete target
ground_truth_trajectory.jsonl	Required artifact; not forwarded to model
metadata.json	Schema and target-contract validation

Label encoding

Raster value	Meaning	Model A class
0	Occupied	2
112	Unknown / unobserved	0
255	Free	1

Unknown is never silently converted to free. For Model B, masked unknown cells define the “guessed” partition while complete GT supplies their occupancy status.



Dataset integrity rules

Scene-grouped split
All sessions from the same asset stay in one split.

Frozen manifest
Session keys, file fingerprints, metadata hashes, and checksum are recorded.

Temporal validity
The target at time t receives frames 0...t only.

Metric target extents remain available to validation metrics such as boundary Chamfer distance; they are not runtime inputs.

4. Live VGGT-Ω Extraction and Normalized Geometry

Backbone execution

- VGGT-Ω weights are loaded once and frozen.
- `torch.no_grad()` prevents backbone gradients.
- CUDA uses BF16 when supported, otherwise FP16.
- The same live forward runs in training and runtime.
- Model A and B reuse one detached extraction.

Extracted tensors

- Patch tokens from layers 4, 11, 17, 23
- Dense depth and confidence
- World-to-camera extrinsics
- Per-frame camera intrinsics, robustly stabilized
- Scene radius and a compact 14-D geometry cue

Scene scale

$$r_{\text{vggt}} = \text{mean}_{X \in P} \|X\|_2$$

Sparse valid VGGT depth samples are back-projected with predicted K and predicted world-to-camera poses, then expressed in the first-camera frame. Their mean radius is the shared normalization unit. Depth and relative translation statistics use the same r_{vggt} .

14-dimensional geometry cue

Components	Count	Definition
Normalized log-depth moments	2	Mean and standard deviation of $\log(D/r)$
Confidence moments	2	Mean and standard deviation of transformed VGGT confidence
Stable intrinsic cue	4	$f_x/W, f_y/H, c_x/W, c_y/H$
Normalized motion magnitudes	3	Mean, maximum, and latest $\ \Delta c\ /r$
Relative rotation	1	Cosine from first-to-latest relative rotation
Latest normalized depth	1	Mean log depth for latest frame
History fraction	1	$N / 34$

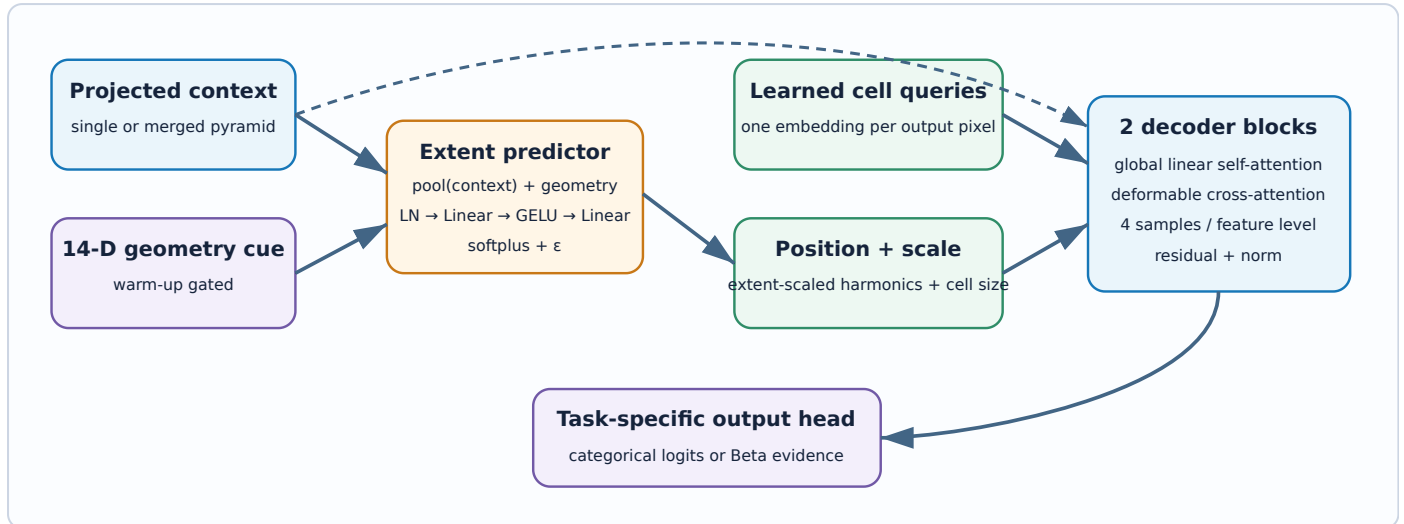
No physical calibration. Camera height is not read or predicted for scale conversion. r_{vggt} is an arbitrary VGGT scene unit. The model is scale-normalized, but it is not metric.

Multi-scale token projection (PDF Equation 24)

$$\tilde{Z} = \text{Concat}_{\ell} (Z^{(\ell)} W_{\ell}), \quad \ell \in \{4, 11, 17, 23\}$$

Each 2048-D cached token level is LayerNorm-projected to 64 channels, reshaped to its patch grid, and spatially resized at 4x, 2x, 1x, and 0.5x. This pyramid is consumed by deformable cross-attention.

5. Direct Method I BEV Decoder



Learned support

$$\bar{e} = \text{softplus}(f_{\text{extent}}(\text{pool}(\tilde{Z}), g)) + 10^{-6}, \quad \Delta = \bar{e} / S$$

The single and merged decoders each predict their own positive extent $\bar{e}$ in scene-radius-normalized VGGT units. Softplus has no upper clamp. S is 512 for the single branch and 800 for the merged branch.

Query construction (implementation of PDF Equation 25)

$$q_p^{(0)} = E_p + \varphi_{\text{pos}}(\bar{e} \cdot \xi_p) + \varphi_{\text{scale}}(\log(1 + \bar{e}), \log(1 + \Delta), g)$$

Each output cell owns a learned embedding. Unit-grid coordinates and sinusoidal harmonics supply position; the first coordinate channels are scaled by the learned extent. The scale projection also receives the normalized geometry cue.

Memory controls

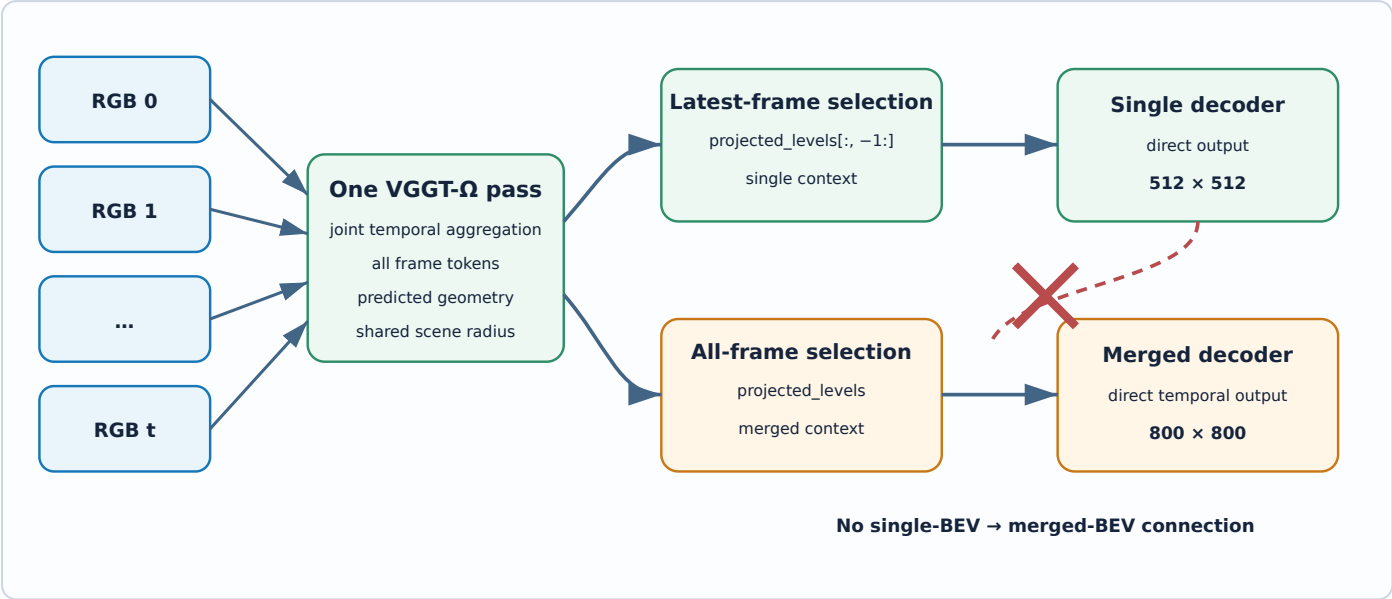
Cell embeddings are stored in chunks; cross-attention queries are chunked; gradient checkpointing is enabled.

Method boundary

Attention samples image-token feature locations. It does not back-project or splat features through predicted depth.

6. Single and Merged Routing

Confirmed current behavior: the merged output is generated as multiple RGB frames → one live VGGT temporal forward → all-frame projected features → merged decoder → merged BEV.



Question	Answer
Is each RGB frame independently converted to a single BEV first?	No. The complete RGB prefix is processed jointly by VGGT-Ω.
Does the merged decoder consume single predictions?	No. It consumes all-frame projected VGGT feature levels.
What supervises merged output?	The cumulative merged GT raster at the same target frame.
Is merged output a persistent online global map?	No. It is the direct prediction for the supplied bounded RGB history.

7. Model A — Observed / Masked BEV

Purpose

Learn the camera-observed area directly. Cells outside established visibility remain the explicit unknown class.

No confidence head is used.

Training targets

- Single masked observed BEV
- Merged masked observed BEV
- Three classes: unknown, free, occupied

Model A output contract

Output	Shape	Meaning
<code>single.class_logits</code>	[B, 3, 512, 512]	Unknown/free/occupied logits
<code>merged.class_logits</code>	[B, 3, 800, 800]	Merged unknown/free/occupied logits
<code>*.predicted_extent_vggt_normalized</code>	[B]	Learned branch support
<code>*.cell_size_vggt_normalized</code>	[B]	Predicted extent divided by raster size

Loss

$$L_{A,s} = \lambda_{CE} \cdot CE_{\text{weighted}} + \lambda_{Dice} \cdot Dice_{\text{occupied}}$$

$$L_A = 0.5 \cdot L_{A,\text{single}} + 0.5 \cdot L_{A,\text{merged}}$$

Configured item	Value
Class order	unknown, free, occupied
Class weights	0.25, 1.0, 2.0
Categorical weight	1.0
Occupied Dice weight	0.25
Single / merged weights	0.5 / 0.5

Inference conversion: apply argmax over the channel axis. Map class 0 → unknown, class 1 → free, and class 2 → occupied. Softmax probabilities may be retained for diagnostics, but Model A is intentionally not trained as a calibrated confidence model.

Isolation from Model B

Model A owns its token projector, learned BEV queries, single decoder, merged decoder, normalization layers, and categorical heads. Its optimizer, AMP scaler, DDP wrapper, loss, and checkpoint are separate. Only the frozen live VGGT extraction is reused.

The production Model-B-only configuration currently disables Model A entirely: it is not instantiated, forwarded, optimized, or checkpointed. The full paired pipeline can re-enable it without changing Model B.

8. Model B — Complete BEV and Evidential Confidence

Purpose

Predict complete occupancy over both observed and unobserved cells, while learning that direct observations should generally carry stronger evidence than guesses.

Training relationship

- Complete BEV supplies occupancy truth.
- Masked BEV partitions observed versus guessed cells.
- No fixed confidence target is supplied.

Beta evidence output

$$\alpha = \text{softplus}(e_{\text{occ}}) + 1, \quad \beta = \text{softplus}(e_{\text{free}}) + 1, \quad S = \alpha + \beta$$

Occupancy mean

$$p_{\text{occ}} = \alpha / S$$

Distribution variance

$$\alpha\beta / [S^2(S+1)]$$

Epistemic uncertainty

$$u = 2 / S$$

Model B output contract

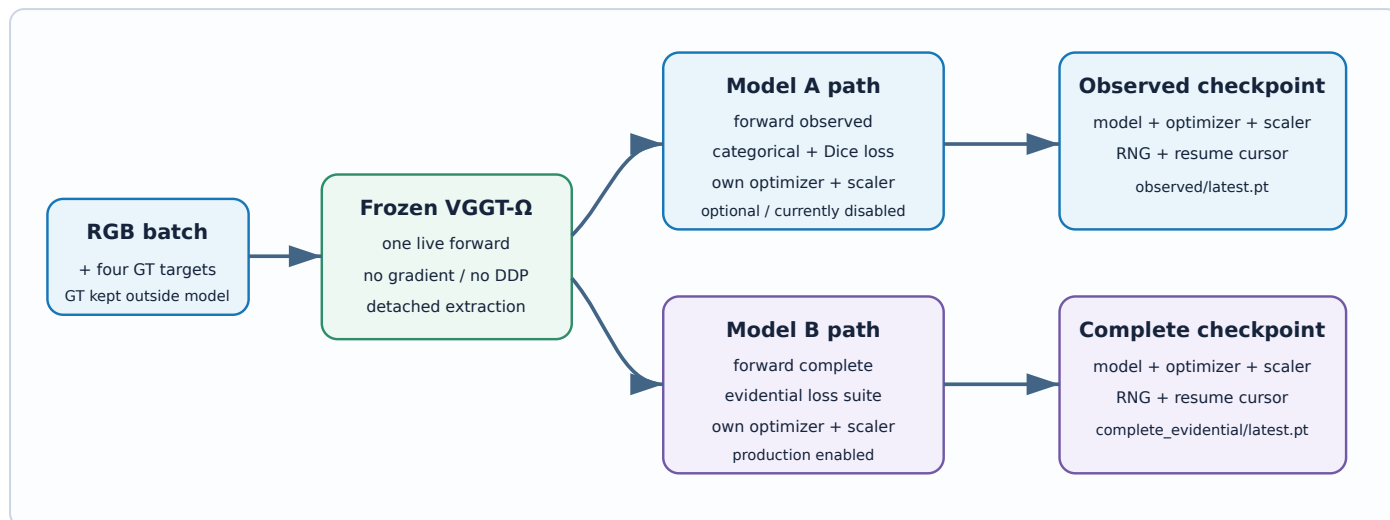
Output per branch	Single shape	Merged shape
raw_evidence	[B,2,512,512]	[B,2,800,800]
alpha_occupied, beta_free	[B,512,512]	[B,800,800]
occupancy_probability	[B,512,512]	[B,800,800]
evidence_strength	[B,512,512]	[B,800,800]
occupancy_distribution_variance	[B,512,512]	[B,800,800]
epistemic_uncertainty	[B,512,512]	[B,800,800]
Extent and cell-size metadata	[B] each	[B] each

Loss components

Term	Role	Weight
Expected Beta NLL	Correct complete occupancy likelihood	1.0
Occupied Dice	Handle sparse obstacle regions	0.25
Wrong-evidence KL	Shrink incorrect class evidence toward Beta(1,1)	0.01
Observation relation	Encourage mean log-strength observed > guessed	0.01
Calibration	Align evidence confidence with correctness, balanced by region	0.05

Confidence is not a post-processing sigmoid. It is encoded in the learned Beta distribution. High S means low epistemic uncertainty; low S means the model retains little evidence. The observation relation is statistical, not a hard rule for each individual cell.

9. Training Execution, DDP, and Checkpoints



Default production schedule

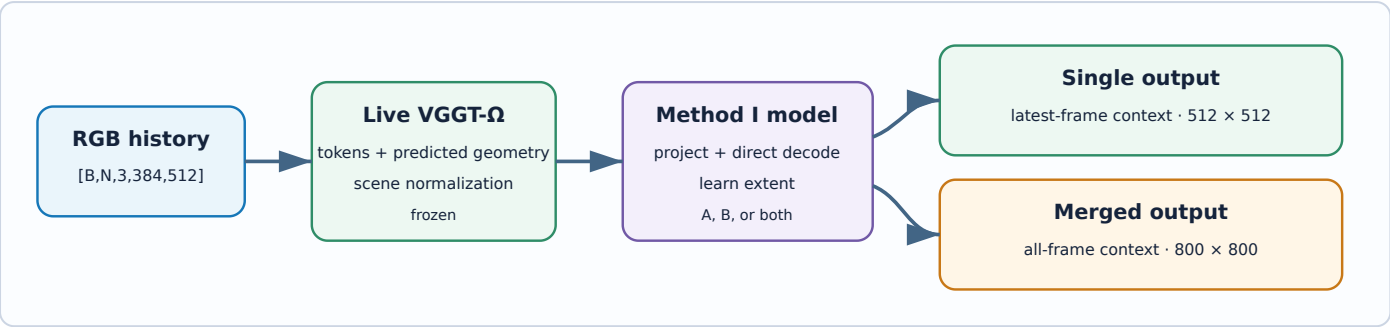
Pipeline	paired_evidential
Enabled models	complete_evidential only
Epochs	10
Per-rank batch	1
Optimizer LR	1×10^{-4}
Weight decay	1×10^{-4}
Geometry cue warm-up / ramp	1 / 2 epochs
Confidence regularizer warm-up / ramp	1 / 2 epochs

Distributed execution

- NCCL/DDP wraps trainable Method I model paths only.
- Frozen 1B VGGT-Ω remains outside DDP buckets.
- Same-NUMA GPU validation occurs before launch.
- Static graph, gradient bucket views, 25 MiB buckets.
- Project runtime requires NCCL 2.30.7 or newer contract.

Checkpoint format version 10 records RGB-only input, learned unbounded normalized extent, 512/800 output sizes, and no metric calibration. Camera-height or fixed-range checkpoints are intentionally incompatible.

10. Runtime Pipeline and Exact I/O



Runtime input contract

Input	Required?	Notes
RGB history	Yes	Contiguous sequence; dimensions divisible by patch size 16
Camera height	No	Not present in forward signature
Camera intrinsics / extrinsics	No	Estimated internally by live VGGT-Ω
Depth	No	Estimated internally by live VGGT-Ω
Trajectory / temporal poses	No	Estimated camera poses are internal; no GT trajectory
BEV range	No	Predicted independently by each decoder

Common diagnostic outputs

Key	Shape / type	Purpose
scene_radius_vggt	[B]	Native VGGT unit used for normalization
estimated_geometry_cue	[B,14]	Scale-free decoder cue
estimated_depth_vggt	[B,N,H,W]	Live predicted depth diagnostic
estimated_intrinsics	[B,N,3,3]	Stabilized sequence K
estimated_camera_from_world_vggt	[B,N,3,4]	Live predicted extrinsics
coordinate_mode	string	vggt_scene_radius_normalized
extent_mode	string	learned_per_sample_unbounded_positive

Runtime is a single call equivalent to `prediction = model(images)`. Target extents, labels, camera height, and simulator parameters do not appear in that call.

11. Coordinate Meaning, Guarantees, and Limitations

How to interpret the learned range

normalized coordinate = native VGGT coordinate / r_{vggt}

branch support = $[-\bar{e}/2, +\bar{e}/2] \times [-\bar{e}/2, +\bar{e}/2]$, cell size = $\bar{e} / S$

The decoder returns $\bar{e}$ directly. Multiplying $\bar{e}$ by r_{vggt} converts it back to the current VGGT model’s arbitrary native scene unit, not to metres.

Guaranteed by implementation

- Extent is positive.
- There is no configured maximum clamp.
- Single and merged predict independent extents.
- Cell size is internally consistent with raster resolution.
- Depth and motion cues share one scene normalization.

Not guaranteed

- One normalized unit is not one metre.
- Predicted extent has no explicit GT extent loss.
- BEV and VGGT point cloud need not agree geometrically.
- Temporal output is not an indefinitely persistent global map.
- Range generalization outside training statistics is unproven.

Identifiability limitation. Occupancy rasters supervise pixel classes, but the dataset does not provide a target expressed in scene-radius-normalized VGGT extent. Therefore $\bar{e}$ is a learned latent scale convention. It receives gradients through positional and scale encoding, but it is not an independently calibrated distance measurement.

Alignment with the source PDF

PDF concept	Current realization	Status
Eq. 16: scene radius	Computed from live predicted depth, K, and pose	Implemented
Eq. 17: joint normalization	Depth and relative motion summarized in one normalized unit	Implemented for model cues
Eq. 24: multi-scale token projection	Layers 4/11/17/23 projected to width 64	Implemented
Eq. 25: learned BEV queries	Learned cell embedding + extent-scaled position + scale cue	Implemented / adapted
Eq. 26–27: self/cross attention	Linear self-attention + deformable cross-attention	Implemented
Eq. 28, 32–33: occupancy decoding/loss	Categorical A and evidential B task heads	Implemented / extended
Eq. 21–23: metric height calibration	Deliberately absent	Non-metric scope
Lift-and-splat geometry	Absent	Preserves Method I boundary

When metric navigation is required

A downstream system must add a physical scale anchor—camera height, wheel odometry, stereo baseline, known object dimension, or calibrated mapping—to interpret BEV cell size in metres. The current checkpoint alone cannot supply that guarantee.

12. Validation, Deployment, and Source Map

Pre-training acceptance checklist

Check	Expected result
Dataset validator	Complete schema-v4 sessions; 6.5 m / 10 m source targets; valid labels
Frozen split	No scene overlap; manifest checksum matches
Model input audit	Only RGB enters <code>forward(images)</code>
Target shapes	Single [B,512,512], merged [B,800,800]
Range outputs	Finite positive learned normalized extent and cell size
Temporal routing	Single uses latest features; merged uses all features
Checkpoint contract	Format v10; no fixed range or metric calibration
GPU preflight	Healthy assigned GPUs, same NUMA, compatible NCCL

Validation metrics

Map quality
Occupied/free IoU, macro F1, boundary Chamfer.

Evidential quality
NLL, strength, uncertainty, calibration by observed/guessed region.

Robustness still needed
Scale shift, temporal flicker, camera changes, latency, cross-head consistency.

Operational sequence

1. Freeze or verify the scene-grouped split manifest.
2. Run the data-only validator and inspect tensor shapes.
3. Run unit tests and a one-step smoke test on an eligible GPU.
4. Launch DDP only on the assigned same-NUMA GPU set.
5. Verify the first runtime report: one VGGT forward, no metric inputs, finite scene radius and extents.
6. Monitor losses and observed-versus-guessed evidence statistics.
7. Validate checkpoints before long runtime evaluation.

Implementation source map

Responsibility	Primary file
Dataset mapping, history, target resizing	<code>src/vggd_bev_method1/data/dataset.py</code>
Batch separation and GT leakage guard	<code>src/vggd_bev_method1/data/collate.py</code>
Live VGGT-Ω and 14-D geometry cue	<code>src/vggd_bev_method1/models/vggd_adapter.py</code>
Token projection and single/merged routing	<code>src/vggd_bev_method1/models/method1.py</code>
Learned range, BEV queries, output heads	<code>src/vggd_bev_method1/models/decoder.py</code>
Self- and cross-attention	<code>src/vggd_bev_method1/models/attention.py</code>
Observed and evidential losses	<code>src/vggd_bev_method1/losses.py</code>
Production training and checkpointing	<code>src/vggd_bev_method1/cli_train_paired.py</code>
Production configuration	<code>configs/remote_model_b_session_prefix_6p5m_10m.toml</code>

Bottom line. This pipeline is an RGB-only-at-runtime, direct Method I system. One frozen live VGGT temporal extraction feeds independent 512² single and 800² merged decoders. The merged decoder consumes all RGB temporal features directly. Model B predicts complete occupancy and a learned Beta evidence distribution; all spatial support is non-metric and expressed in learned scene-radius-normalized VGGT units.